

QUICKBORN, Germany

WMO: 101460

Lat: 53.7331N

Lon: 9.8775E

Elev: 12

StdP: 101.18

Time Zone: 1.00 (EUC)

Period: 96-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.7	-7.4	-12.5	1.3	-7.8	-9.7	1.6	-5.6	10.9	7.6	9.8	6.9	1.6	80	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	28.3	18.9	26.2	18.5	24.3	17.7	20.4	25.6	19.4	24.1	18.5	22.7	3.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.5	21.9	17.7	12.7	20.9	16.9	12.0	20.0	58.6	25.8	55.2	24.2	52.4	22.8	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.5	6.6	-13.5	32.1	4.7	1.9	-16.9	33.5	-19.6	34.5	-22.3	35.6	-25.6	36.9		
(5)			-13.7	22.4	4.6	0.7	-17.1	22.9	-19.8	23.3	-22.4	23.7	-25.7	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	1.5	2.1	4.2	8.4	12.1	14.8	17.2	17.0	13.8	9.6	5.2	1.7		
	DBStd	6.80	4.69	4.15	3.98	3.79	3.34	3.19	2.92	2.77	3.00	3.34	3.75	4.49		
	HDD10.0	1216	262	223	183	75	16	1	0	0	3	48	148	257		
	HDD18.3	3461	521	456	439	298	195	114	58	61	141	270	394	514		
	CDD10.0	853	0	1	2	27	82	146	223	217	116	36	3	1		
	CDD18.3	56	0	0	0	0	2	9	23	19	3	0	0	0		
	CDH23.3	622	0	0	0	4	33	104	268	179	32	1	0	0		
	CDH26.7	142	0	0	0	0	5	22	72	41	2	0	0	0		
(14) Wind	WSAvg	3.0	3.3	3.6	3.5	3.2	2.9	2.8	2.7	2.5	2.4	2.9	2.9	3.1		
(15) Precipitation	PrecAvg	797	67	56	56	41	60	77	86	80	69	63	68	75		
	PrecMax	1090	137	148	118	83	154	163	206	187	204	186	140	174		
	PrecMin	521	0	9	8	2	11	11	27	31	26	27	8	24		
	PrecStd	130	35	30	30	20	29	39	46	37	40	32	33	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	12.3	16.9	23.6	26.7	29.3	31.7	30.1	26.6	19.6	14.7	12.1		
		MCWB	10.1	9.4	11.2	14.8	17.8	20.1	19.7	19.3	18.2	16.0	12.5	10.6		
	2%	DB	10.2	10.5	14.0	20.4	23.6	26.4	28.6	27.5	23.9	17.4	12.9	10.3		
		MCWB	8.8	8.8	9.6	13.0	16.0	18.3	19.1	18.9	17.5	14.4	11.4	9.2		
	5%	DB	8.8	9.2	11.7	18.0	21.3	23.7	26.3	25.2	21.4	15.9	11.6	8.9		
		MCWB	7.6	7.9	9.1	11.9	14.7	17.2	18.7	18.3	16.2	13.7	10.3	7.8		
	10%	DB	7.6	7.8	10.0	15.5	19.1	21.5	24.1	23.1	19.3	14.7	10.4	7.7		
		MCWB	6.6	6.7	7.7	10.9	13.8	16.1	17.8	17.7	15.4	13.0	9.1	6.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	10.5	12.2	15.9	19.1	21.2	21.7	21.6	19.5	16.6	13.6	11.3		
		MCDB	11.5	11.4	15.0	21.0	24.5	26.9	27.3	27.0	24.1	19.0	14.3	12.1		
	2%	WB	9.1	9.2	10.7	13.8	17.2	19.5	20.4	20.0	18.1	15.1	11.7	9.3		
		MCDB	10.1	10.1	12.7	18.7	21.8	24.5	25.9	24.6	22.1	16.7	12.7	10.1		
	5%	WB	7.7	8.0	9.4	12.6	15.7	18.1	19.4	19.1	17.0	14.1	10.4	8.0		
		MCDB	8.5	9.2	11.5	16.8	19.5	22.0	24.6	23.0	20.3	15.5	11.4	8.6		
	10%	WB	6.7	6.7	8.0	11.3	14.3	16.8	18.4	18.2	16.0	13.1	9.2	6.9		
		MCDB	7.5	7.7	9.6	14.9	17.9	20.3	22.9	21.9	18.6	14.5	10.1	7.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.3	7.5	10.1	10.6	10.2	10.2	10.0	9.3	7.2	5.1	4.5		
		MCDBR	4.9	6.0	10.9	14.4	14.9	14.3	15.0	13.7	12.7	8.2	5.5	4.8		
	5% WB	MCWBR	4.4	4.6	7.4	7.9	8.2	7.4	6.8	6.5	7.0	5.8	4.4	4.2		
		MCWBR	4.7	5.3	9.3	12.2	12.6	12.5	13.0	11.3	10.9	7.5	5.2	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.337	0.367	0.419	0.411	0.409	0.428	0.421	0.393	0.355	0.333	0.310		
		taud	2.340	2.345	2.291	2.178	2.243	2.276	2.240	2.271	2.333	2.426	2.386	2.334		
	Ebn at Noon		642	754	807	806	836	841	815	797	772	726	619	573		
		Edh at Noon	58	79	102	130	128	126	128	118	98	73	56	49		
(44) All-Sky Solar Radiation	RadAvg		0.56	1.22	2.43	4.03	5.07	5.31	5.03	4.15	3.01	1.65	0.70	0.41		
	RadStd		0.07	0.17	0.24	0.50	0.57	0.41	0.67	0.31	0.34	0.19	0.07	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-143	N/A	N/A

Nomenclature: See separate page